

Storage Devices — Detailed Lecture Notes

JKSSB Computer Awareness

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10.1 The storage hierarchy

Storage is organized into levels according to how close it sits to the processor and what job it does. The CPU works directly with primary storage; secondary storage holds data permanently until it is needed and feeds it to main memory; tertiary storage is an archival level used for large, rarely accessed collections.

Definition (Primary storage). *The directly CPU-accessible working memory of a computer, such as RAM, ROM, and cache; it holds the data and instructions currently in use.*

Definition (Secondary storage). *Non-volatile storage that retains data when the power is off and is not accessed directly by the CPU — for example a hard disk, SSD, optical disc, or USB flash drive.*

Definition (Tertiary storage). *An archival storage level for large volumes of infrequently used data, typically robotic tape libraries whose media are loaded on demand.*

Level	Examples	Key property
Primary	RAM, ROM, cache	Directly accessible by the CPU
Secondary	HDD, SSD, optical disc, USB drive	Non-volatile; holds files
Tertiary	Robotic tape library	Archival; loaded on demand

The CPU does not read a file straight from secondary storage. The operating system brings the required data into main memory first, and the CPU then works with that copy. This is why the hierarchy matters for an exam scenario: a large program on disk is limited by the memory and storage that actually feed it, not only by processor speed.

10.2 Magnetic storage

Magnetic storage records data by magnetizing a surface, and it is read back by detecting those magnetic patterns. Hard disks and magnetic tape are the two common magnetic media, but they differ sharply in how they are reached.

10.2.1 Hard disk drive

A hard disk drive (HDD) stores data on *rotating platters* coated with a magnetic material. A *read/write head* mounted on an actuator arm floats just above the platter surface and reads or

writes as the platter spins. The head moves inward and outward to reach different tracks. Because the platters spin and the arm moves, an HDD is a *mechanical* device with moving parts.

Definition (Hard disk drive (HDD)). *Magnetic secondary storage that stores data on rotating platters read and written by a moving read/write head; it is mechanical and contains moving parts.*

10.2.2 Magnetic tape

Magnetic tape stores data on a long, thin magnetic ribbon wound in a reel or cartridge. The tape is read or written in order, position by position, so a record far from the current position takes longer to reach. This makes tape a *sequential-access* medium. It is inexpensive and dense, which is why it is used for backup and long-term archival rather than for routine access.

Definition (Magnetic tape). *Magnetic storage on a long tape that is read and written in order; it is a sequential-access medium used mainly for backup and archival.*

10.3 Optical storage

Optical storage records data as *pits and lands* — tiny physical markings — on a disc and reads them with a *laser*. The reflected light pattern encodes the data. Optical discs are removable and non-volatile, so they are convenient for distributing and archiving files.

The three common families are the compact disc (CD), the digital versatile disc (DVD), and the Blu-ray disc. Blu-ray uses a shorter-wavelength laser and tighter tracks, allowing it to store more than a DVD, which in turn stores more than a CD.

Definition (Optical storage). *Storage in which a laser reads data encoded as pits and lands on a disc, such as a CD, DVD, or Blu-ray disc.*

Medium	Approximate capacity (single layer)	Reading method
CD	700 MB	Laser
DVD	4.7 GB	Laser (shorter wavelength)
Blu-ray	25 GB	Laser (shortest wavelength)

Order the capacities from smallest to largest as $CD < DVD < \text{Blu-ray}$. A common test error is to swap the middle entries or to reverse the order entirely; fix the sequence by remembering that CD is the oldest and smallest and Blu-ray is the newest and largest.

Example 10.3.1. *An item asks which optical disc holds the most data. Compare the standard single-layer figures: $CD \approx 700 \text{ MB}$, $DVD \approx 4.7 \text{ GB}$, and $\text{Blu-ray} \approx 25 \text{ GB}$. Since $25 \text{ GB} > 4.7 \text{ GB} > 700 \text{ MB}$, the Blu-ray disc holds the most.*

10.4 Flash and solid-state storage

Flash memory stores data in semiconductor cells that hold their state without power. There are no rotating platters and no moving heads, so flash devices are *solid-state*. Common examples are the solid-state drive (SSD), the USB flash drive, and the memory card.

Definition (Flash memory). *Non-volatile semiconductor storage that keeps data in cells with no moving parts; the technology behind SSDs, USB flash drives, and memory cards.*

Definition (Solid-state drive (SSD)). *A flash-based storage device with no moving parts. It is faster and more shock-resistant than a mechanical hard disk and has a high transfer rate.*

Because nothing spins and no head has to move, flash can start a transfer almost immediately and is far less sensitive to being bumped or dropped. Among a hard disk drive, an optical drive, magnetic tape, and an SSD, the **SSD** has the highest transfer rate.

10.5 Cloud storage

Cloud storage keeps data on remote servers in a data centre and makes it available over the Internet. The user sees a folder or an application, while the provider manages the underlying hardware. Even though the storage feels abstract, it ultimately rests on physical secondary storage — arrays of disks and solid-state drives — on the provider’s machines.

Definition (Cloud storage). *Storage in which data is held on remote servers and accessed over the Internet rather than on a locally attached device.*

Cloud storage is not main memory and it is not the same as primary storage. It is remote secondary storage reached through a network connection, useful for backup, sharing, and access from several devices.

10.6 Hard disk versus solid-state drive

The comparison below is the most frequently tested near-neighbour pair in this topic. The hard disk is magnetic and mechanical; the SSD is flash and solid-state. Keep the technology and the consequences together.

Aspect	Hard disk drive (HDD)	Solid-state drive (SSD)
Technology	Magnetic	Flash memory
Moving parts	Yes — spinning platters, moving head	None
Speed	Usually slower transfer rate	Higher transfer rate (fastest here)
Shock resistance	More vulnerable to shock	More shock-resistant
Typical use	Bulk data at low cost per GB	Fast system and boot drive

Example 10.6.1. *A candidate reads “the fastest storage device is the hard disk.” That is incorrect. The hard disk is magnetic and mechanical, with a spinning platter and a moving head. The SSD is flash-based with no moving parts and has the higher transfer rate. The correct choice is the SSD (KB PYQ-27).*

10.7 Access methods: random and sequential

An access method describes how quickly a device can reach a stored location.

Definition (Random (direct) access). *An access method in which any location can be reached in approximately equal time; RAM and hard disks behave this way.*

Definition (Sequential access). *An access method in which locations are reached in order, so the time depends on how far the target is from the current position; magnetic tape works this way.*

Access method	Behavior	Examples
Random (direct)	Any location in about equal time	RAM, hard disk
Sequential	Reached in order	Magnetic tape

Memory and disks are approximately random access because the electronics or the head can be directed to a location without stepping through everything before it. Tape is sequential because the ribbon must physically pass over the read head in order.

10.8 Non-volatility of storage

Non-volatility is the property that stored contents survive when the power is switched off. Storage devices are designed to be non-volatile so that files remain available after shutdown. Primary storage such as RAM is *volatile*: its contents disappear when power is removed.

Definition (Non-volatile storage). *Storage that retains its contents without power; hard disks, SSDs, optical discs, and USB drives are non-volatile.*

The one-line rule to carry into the exam is: RAM is volatile; storage is non-volatile. This is exactly why unsaved work in RAM can be lost in a power failure while a file already saved to disk remains.

10.9 Exam retrieval and common confusions

Use a two-direction recall drill. Given a device, state its technology and its access method; given a technology or capacity, state the matching device.

- HDD $\leftrightarrow$ magnetic, mechanical, random access.
- Magnetic tape $\leftrightarrow$ magnetic, sequential access, backup.
- CD / DVD / Blu-ray $\leftrightarrow$ optical, capacities 700 MB / 4.7 GB / 25 GB (single layer).
- SSD / USB drive / memory card $\leftrightarrow$ flash, no moving parts; the SSD has the highest transfer rate.
- Cloud storage $\leftrightarrow$ remote servers over the Internet.
- RAM volatile versus storage non-volatile.

Three distinctions prevent most errors in this topic:

1. Do not call the SSD magnetic or mechanical. The magnetic and mechanical device is the hard disk; the SSD is flash with no moving parts.
2. Do not claim that the hard disk has the highest transfer rate; the SSD does (KB PYQ-27).
3. Do not reverse the optical ordering. It is CD < DVD < Blu-ray for standard single-layer capacities, and do not confuse volatile RAM with non-volatile storage.

Example 10.9.1. *An item asks which device stores data without any moving parts. Eliminate the hard disk (spinning platters and a moving head) and magnetic tape (a moving ribbon). An optical drive also spins its disc. The flash-based SSD stores data in semiconductor cells with no moving parts, so it is the answer.*

Exercises

1. Distinguish primary, secondary, and tertiary storage and give one example of each.
2. Explain why an SSD is described as having no moving parts, and why it is more shock-resistant than a hard disk.
3. Order CD, DVD, and Blu-ray by single-layer capacity from smallest to largest, with the approximate figures.
4. Classify RAM, a hard disk, and magnetic tape as random access or sequential access.
5. A student writes: “RAM is non-volatile, and the hard disk is the fastest storage device.” Correct both errors.

Solutions

1. Primary storage is directly accessible by the CPU (for example RAM). Secondary storage is non-volatile and holds data when power is off (for example a hard disk). Tertiary storage is archival storage loaded on demand (for example a robotic tape library).
2. An SSD stores data in flash semiconductor cells, with no spinning platter and no moving read/write head. With no moving parts to be disturbed, it tolerates shock better than a hard disk and transfers data faster.
3. CD $\approx$ 700 MB, then DVD $\approx$ 4.7 GB, then Blu-ray $\approx$ 25 GB, for single-layer media.
4. RAM is random access; a hard disk is approximately random access; magnetic tape is sequential access.
5. RAM is *volatile*, not non-volatile: its contents are lost without power. The device with the highest transfer rate is the *SSD*, not the hard disk; the hard disk is magnetic and mechanical with moving parts.